

7.1.

Let $\{f_n: X \rightarrow Y\}$ be a uniformly convergent sequence of bounded functions.

Prove it is uniformly bounded.

Proof. Given $\epsilon > 0$, there is $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow \sup_x |f_n - f| \leq \epsilon$, where f is the limit function of f_n .

Therefore, $\sup_x |f| \leq \sup_x |f_N| + \epsilon$, the limit function f is bounded. Now,

for each $1 \leq i < N$, let M_i be a bound of f_i . Then, for all $k \in \mathbb{N}$,

$$\sup |f_k| \leq \max\{M_1, \dots, M_{N-1}, \sup |f| + \epsilon\}.$$

7.2-3. Let $f_n, g_n: X \rightarrow Y$ be functions.

1). If $\{f_n\}, \{g_n\}$ converges uniformly, then $\{f_n + g_n\}$ converges uniformly.

2). If $\{f_n\}, \{g_n\}$ converges uniformly of bounded functions, then $\{f_n g_n\}$ converges uniformly.

3). Find Counter example for 2), the boundedness omitted.

Proof. 1) can be shown directly, by $\sup |f_n + g_n| \leq \sup |f_n| + \sup |g_n|$.

But 2) is needed the boundedness condition for example,

$$f_n: [0, 1] \rightarrow \mathbb{R}: x \mapsto \frac{n}{n+1} \cdot \frac{1}{x} \quad \left(\sup |f_n - \frac{1}{x}| = \frac{1}{n+1} \rightarrow 0 \right)$$

$$g_n: [0, 1] \rightarrow \mathbb{R}: x \mapsto \frac{n+1}{n^2} \cdot \frac{1}{x} \quad \left(\sup |g_n - \frac{1}{x}| = \frac{1}{n} + \frac{1}{n^2} \rightarrow 0 \right)$$

But $f_n g_n(x) = \frac{1}{nx}$ is not uniformly convergent, because $\sup |f_n g_n - 0| = \infty$, but $f_n g_n \rightarrow 0$.

Now suppose that f_n and g_n are bounded (of course, uni. convergent.)

By the above problem, f_n and g_n are uniformly bounded. Now,

$$|f_n g_n - f g| = |(f_n - f) \cdot g_n + (g_n - g) \cdot f| \leq |g_n| |f_n - f| + |f| |g_n - g|$$

gives the result, (More specifically, given $\epsilon > 0$, there is $N \in \mathbb{N}$ s.t.

$$n \geq N \Rightarrow \sup |f_n - f|, \sup |g_n - g| < \epsilon.$$

Therefore, for these $n \geq N$,

$$\sup |f_n g_n - f g| \leq 2M\epsilon, \text{ where } M \text{ is bound of } |g_n| \text{ and } |f|.$$

7.4. Consider a map defined on Real Numbers

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{1+n^2x}$$

- 1). Find $x \in \mathbb{R}$ for the series Converges absolutely
- 2) Find interval for the series Converges Uniformly
- 3) Find interval for the Converges Uniformly fails.
- 4). If the series converges, is f continuous? $\exists b \in \frac{1}{2}d \in \mathbb{R}$
- 5). " , f bounded?

Sol). Since $\left| \frac{1}{1+n^2x} \right| \leq \frac{1}{n^2c}$ if $x > 0$, for all $x > 0$, it Converges absolutely.

75. Let

$$f_n(x) = \begin{cases} 0 & x \notin \left[\frac{1}{n+1}, \frac{1}{n}\right] \\ \sin^2 \frac{\pi}{x} & x \in \left[\frac{1}{n+1}, \frac{1}{n}\right] \end{cases}$$

Show that $\{f_n\}$ converges to a continuous map, but not uniformly.

Observe that: $x \in \left[\frac{1}{n+1}, \frac{1}{n}\right] \Leftrightarrow \frac{\pi}{x} \in [n\pi, (n+1)\pi]$

First, for any $x_0 \in \mathbb{R}$, $f_n(x_0) \rightarrow 0$ as $n \rightarrow \infty$. (If $x_0 \leq 0$, then clearly $f_n(x_0) = 0$, and if $x_0 > 0$, then for some $N \in \mathbb{N}$, $n \geq N \Rightarrow x_0 > \frac{1}{n} > \frac{1}{n+1} > 0$. Thus $f_n(x_0) = 0, \forall n \geq N$.)

But, for any $n \in \mathbb{N}$, $\sup |f_n(x) - 0| = 1 \neq 0$, thus it is not uniformly.

Meanwhile, see that

$$\sum_{n=1}^{\infty} f_n = \begin{cases} 0 & x \notin (0, 1] \\ \sin^2 \frac{\pi}{x} & x \in (0, 1] \end{cases}$$

Clearly, it converges absolutely, but not uniformly, because for all $N \in \mathbb{N}$,

$$\sup \left| \sum_{n=1}^{\infty} f_n - \sum_{i=1}^N f_i \right| = 1. \quad \square$$

16. Prove that the series

$$\sum_{n=1}^{\infty} (-1)^n \cdot \frac{x^2 + n}{n^2}$$

Converges uniformly in any bounded interval but not absolutely for any x .

Proof. First, since $\sum_{n=1}^{\infty} (-1)^n$ is bounded and $\frac{x^2 + n}{n^2} \rightarrow 0$ as $n \rightarrow \infty$ for any x , mathematically

the Dirichlet's Test gives the series Converges. But Not absolutely:

$$\sum_{n=1}^{\infty} \frac{x^2 + n}{n^2} = \sum_{n=1}^{\infty} \frac{x^2}{n^2} + \frac{1}{n} \quad \text{diverges.}$$

Nevertheless, it Converges uniformly, because: for fixed interval $[a, b]$ and for any $N \in \mathbb{N}$,

$$\left| \sum_{n=1}^{\infty} (-1)^n \cdot \frac{x^2 + n}{n^2} - \sum_{n=1}^N (-1)^n \cdot \frac{x^2 + n}{n^2} \right| = \left| \sum_{n=N+1}^{\infty} (-1)^n \cdot \frac{x^2 + n}{n^2} \right| \stackrel{(*)}{\leq} \left| \sum_{n=N+1}^{\infty} (-1)^n \cdot \frac{(\max_{[a,b]} x)^2 + n}{n^2} \right| \rightarrow 0$$

therefore, Converges uniformly. (The inequality $(*)$ is given by difference of two adjacent terms)

17. Define a map on $\mathbb{R}$ as

$$f_n(x) = \frac{x}{1 + nx^2}$$

1. $\{f_n\}$ Converges uniformly.

2. $f'(x) = \lim_{n \rightarrow \infty} f'_n(x)$ except $x=0$.

$$\text{Proof. } f'_n(x) = \frac{(1 + nx^2) - 2nx^2}{(1 + nx^2)^2} = \frac{1 - nx^2}{(1 + nx^2)^2} = 0 \quad \text{if } x = \pm \frac{1}{\sqrt{n}}$$

$$f_n\left(\frac{1}{\sqrt{n}}\right) = \frac{1}{2\sqrt{n}} \rightarrow 0 \quad \text{as } n \rightarrow \infty \quad \text{clearly, therefore}$$

$$\sup |f_n| \leq \frac{1}{2\sqrt{n}} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

$$\text{And, for } x \neq 0, \quad f'_n(x) = \frac{1 - nx^2}{(1 + nx^2)^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

$$\text{But } f'_n(0) = 1, \quad \text{while } f'(0) = 0.$$

78. Let $\{x_n\}$ be a sequence of distinct points of (a, b) , and if $\sum |C_n|$ converges, then

$$f(x) = \sum_{n=1}^{\infty} C_n \cdot I(x - x_n) \quad (\text{as } x \leq b)$$

Converges uniformly and f is continuous except $\{x_n\}$.

Proof. Converges uniformly is given by the Weierstrass Test, directly.

79. Let $\{f_n\}$ be a sequence of continuous functions which converges uniformly to f on E .

Prove that the statement:

If $\{x_n\}$ in E converges to $x \in E$, then $\lim_{n \rightarrow \infty} f_n(x_n) = f(x)$.

Proof. Given $\epsilon > 0$, there exists $N_1 \in \mathbb{N}$ s.t. $n \geq N_1 \Rightarrow \sup_E |f_n - f| \leq \frac{\epsilon}{2}$.

And, since f_n is continuous, there is $N_2 \in \mathbb{N}$ s.t. $m \geq N_2 \Rightarrow |f_n(x_m) - f_n(x)| \leq \frac{\epsilon}{2}$.

Now, for $n \geq \max\{N_1, N_2\}$,

$$|f_n(x_n) - f(x)| \leq |f_n(x_n) - f_n(x)| + |f_n(x) - f(x)| \leq \epsilon.$$

Converse?

10. Denote $\{x\}$ as the fractional part of the real number x . Consider the function defined on $\mathbb{R}$

$$f(x) = \sum_{n=1}^{\infty} \frac{\{nx\}}{n^2}$$

Find all discontinuities of f , and show that they form a countable dense set.

Show that f is Riemann-integrable on every bounded interval.

Proof. Since $\left| \frac{\{nx\}}{n^2} \right| \leq \frac{1}{n^2}$, f converges uniformly, by the Weierstrass Theorem.

Observe that: f is continuous at x iff $f(x) = f(x^+) = f(x^-)$.

Hence that $f_N(x) = \sum_{n=1}^N \frac{\{nx\}}{n^2}$. Since $f_N \rightarrow f$ uniformly, for any $x \in \mathbb{R}$,

$$f(x^-) = \lim_{t \rightarrow x^-} f(t) = \lim_{N \rightarrow \infty} \lim_{t \rightarrow x^-} f_N(t) = \lim_{N \rightarrow \infty} f_N(x^-) \quad (\text{any } N \in \mathbb{N} \text{ and } x \in \mathbb{R}, \lim_{t \rightarrow x^-} f_N(t) \text{ exists,}$$

and similarly $f(x^+) = \lim_{t \rightarrow x^+} f(t) = \lim_{N \rightarrow \infty} f_N(x^+) = \lim_{N \rightarrow \infty} f_N(x) = f(x)$. because it has only finitely many simple discontinuity in any neighborhood. Moreover,

Let $x \in \mathbb{R} \setminus \mathbb{Q}$, irrational. Then, for any $N \in \mathbb{N}$, $\lim_{t \rightarrow x^+} f_N(t) = f_N(x)$, because $\{nx^+\} = \{nx\}$.

$\lim_{t \rightarrow x^-} f_N(t) = f_N(x)$, because the possible discontinuity of f_N only occurs on the set

$$\left\{ \frac{k}{m} \mid k \in \mathbb{Z}, m=1, 2, \dots, N \right\}.$$

Therefore $f(x^-) = \lim_{N \rightarrow \infty} f_N(x^-) = \lim_{N \rightarrow \infty} f_N(x) = f(x)$.

Let $x \in \mathbb{Q}$. Denote $x = \frac{p}{q}$, as rational form. Then, for all $N \geq p$,

$$f_N(x^-) - f_N(x) \geq \frac{1}{p^2}, \text{ therefore } f(x^-) \neq f(x).$$

But, since each $f_N \in \mathcal{R}$ because f_N has only finitely many discontinuity on bounded interval and $f_N \rightarrow f$ uniformly, $f \in \mathcal{R}$. $\square$

11. Suppose $\{f_n\}, \{g_n\}$ are defined on E and

1) $\sum f_n$ has uniformly bounded partial sums.

2) $g_n \rightarrow 0$ uniformly on E

3) $g_1 \geq g_2 \geq \dots$

Then, $\sum f_n g_n$ Converges uniformly.

Proof. Let $M > 0$ be a bound of $\{\sum f_n\}$. That is, for all $m \in \mathbb{N}$, $|\sum_{k=1}^m f_k| < M$.

Let $x \in E$ be given. By the Abel's formula,

$$\sum_{k=n}^m f_k(x) g_k(x) = \left(\sum_{k=1}^m f_k(x) \right) \cdot g_m(x) - \left(\sum_{k=1}^{n-1} f_k(x) \right) \cdot g_n(x) - \sum_{k=n}^{m-1} \left(\sum_{i=1}^k f_i(x) \right) \cdot (g_{k+1}(x) - g_k(x))$$

Now, for $m > n \geq N$

$$\left| \sum_{k=n}^m f_k(x) g_k(x) \right| \leq \left| \left(\sum_{k=1}^m f_k(x) \right) \cdot g_m(x) \right| + \left| \left(\sum_{k=1}^{n-1} f_k(x) \right) \cdot g_n(x) \right| + \left| \sum_{k=n}^{m-1} \left(\sum_{i=1}^k f_i(x) \right) \cdot (g_{k+1}(x) - g_k(x)) \right|$$

$$\leq M \cdot |g_m(x)| + M |g_n(x)| + M \cdot \left| \sum_{k=n}^{m-1} (g_{k+1}(x) - g_k(x)) \right|$$

$$= M \cdot |g_m(x)| + M |g_n(x)| + M |g_m(x) - g_n(x)|$$

$$= 2g_m(x) \rightarrow 0 \text{ as } N \rightarrow \infty$$

Since g_m Converges uniformly, so is $\sum f_n g_n$. $\square$

7.12. Let $g, f_n: (0, \infty) \rightarrow \mathbb{R}$ be Riemann-integrable on $[t, T]$ whenever $0 < t < T < \infty$.
 If $|f_n| < g$, and $f_n \rightarrow f$ uniformly on every compact subset of $(0, \infty)$, and $\int_0^\infty g(x) dx < \infty$,

then $\lim_{n \rightarrow \infty} \int_0^\infty f_n(x) dx = \int_0^\infty \lim_{n \rightarrow \infty} f_n(x) dx$.

Proof. First, $\lim_{t \rightarrow 0} \lim_{T \rightarrow \infty} \int_t^T f_n(x) dx \leq \lim_{t \rightarrow 0} \lim_{T \rightarrow \infty} \int_t^T |f_n(x)| dx \leq \int_0^\infty g(x) dx < \infty$,
 for any $n \in \mathbb{N}$, $\int_0^\infty f_n(x) dx$ converges.

Since $f_n \rightarrow f$ uniformly, for any $t < T$, f is Riemann-integrable on $[t, T]$ with

$$\lim_{n \rightarrow \infty} \int_t^T f_n(x) dx = \int_t^T f(x) dx.$$

Moreover, $\int_t^T f_n \rightarrow \int_t^T f$ uniformly on $(0, T)$, because

$$\sup_{t \in (0, T)} \left| \int_t^T f_n - \int_t^T f \right| \leq \sup_{t \in (0, T)} \int_t^T |f_n - f| \leq \sup_{t \in (0, T)} |f_n(t) - f(t)| \cdot T \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Since $0 \in \mathbb{R}$ is a limit point of $(0, T)$,

$$\lim_{t \rightarrow 0} \lim_{n \rightarrow \infty} \int_t^T f_n(x) dx = \lim_{n \rightarrow \infty} \lim_{t \rightarrow 0} \int_t^T f_n(x) dx, \text{ There is, } \int_0^T f(x) dx = \lim_{n \rightarrow \infty} \int_0^T f_n(x) dx.$$

$$\text{And, similarly, } \int_0^\infty f(x) dx = \lim_{T \rightarrow \infty} \lim_{n \rightarrow \infty} \int_0^T f_n(x) dx \stackrel{(*)}{=} \lim_{n \rightarrow \infty} \lim_{T \rightarrow \infty} \int_0^T f_n(x) dx = \lim_{n \rightarrow \infty} \int_0^\infty f_n(x) dx. \quad \square$$

Rigorously Comment for $(*)$: for each $n \in \mathbb{N}$, $\lim_{T \rightarrow \infty} \int_0^T f_n(x) dx$ exists, take this to A_n .

That is, $A_n := \int_0^\infty f_n(x) dx$. And, To show $\int_0^\infty f(x) dx$ exists, we show that $\int_0^\infty |f|$.
 Suppose that $\lim_{T \rightarrow \infty} \int_0^T |f(x)| dx$ does Not exist. (The existence of $\int_0^T |f|$ can be proven, similarly above)

Since $\int_0^T |f(x)| dx$ increase, the only possibility is that it diverges.

$$\text{Take } R > 0 \text{ s.t. } \int_0^R |f(x)| dx > \int_0^\infty g(x) dx.$$

Since $f_n \rightarrow f$ uniformly on $[t, R]$, for any $0 < t < R$, we can take $N \in \mathbb{N}$ s.t.

$$\int_0^R |f(x)| dx > \int_0^R |f_n(x)| dx > \int_0^\infty g(x) dx \quad (1)$$

$$\text{or } \int_0^R |f_n(x)| dx > \int_0^R |f(x)| dx > \int_0^\infty g(x) dx \quad (2)$$

Either (1) or (2), Contradiction. Thus $\int_0^\infty |f(x)| dx$ converges, so is $\int_0^\infty f(x) dx$.

The remain can be proved same way, Thm. 7.11.

17, 13.

Let $\{f_n\}$ be a sequence of monotone increasing maps on $\mathbb{R}$ with $0 \leq f_n \leq 1$ for all $n \in \mathbb{N}$.

a). Helly's selection Theorem.

There is a function f and subsequence $\{n_k\}$ s.t.

$$f(x) = \lim_{k \rightarrow \infty} f_{n_k}(x) \quad \text{for all } x \in \mathbb{R}.$$

b). Moreover, 'if f in a) is continuous, then $f_{n_k} \rightarrow f$ uniformly on Compact sets.

Proof. Denote $\mathbb{Q} = \{x_n | n \in \mathbb{N}\}$. We will use the fact that:

Every bounded sequence in a Compact set has a convergent subsequence.

First, since $\{f_n(x_1)\}$ is bounded in $[0, 1]$, it has a convergent subsequence.

Put the corresponding maps as $\{f_{1,k}\}$. That is, $\{f_{1,k}(x_1)\}$ converges, $(1,k) < (1,k+1)$ for all $k \in \mathbb{N}$.

Second, similarly, $\{f_{1,k}(x_2)\}$ has a convergent subsequence, put $\{f_{2,k}\}$.

Inductively, for any $m \in \mathbb{N}$,

$$\{(m,k)\} \subseteq \{(m-1,k)\} \subseteq \dots \subseteq \{(2,k)\} \subseteq \{(1,k)\}$$

Now, consider an array

s_0	f_1	f_2	f_3	$\dots$
s_1	$f_{1,1}$	$f_{1,2}$	$f_{1,3}$	$\dots$
s_2	$f_{2,1}$	$f_{2,2}$	$f_{2,3}$	$\dots$
$\vdots$				

Given $x_n \in \mathbb{Q}$, $f_{1,k}(x_n)$ converges, because $f_{k,k}$ ($k \geq n$) is a subsequence of $f_{n,k}$ and has hereditary order. Now, we obtain a subsequence $\{f_{n_k}\}$, which converges on $\mathbb{Q}$. (c)

define a map $\hat{f}: \mathbb{Q} \rightarrow \mathbb{R}; x \mapsto \lim_{k \rightarrow \infty} f_{n_k}(x)$, and

$$f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \sup_{\substack{r \leq x \\ r \in \mathbb{Q}}} \hat{f}(r)$$

Suppose that f is continuous at $x \in \mathbb{R}$. Then, for a rational numbers sequence $\{r_n\}$ s.t. $r_n \leq x, r_n \nearrow x$,

$$f(x) = \sup_{t \leq x} f(t) = \lim_{n \rightarrow \infty} f(r_n) = \lim_{n \rightarrow \infty} \hat{f}(r_n) = \lim_{n \rightarrow \infty} \lim_{k \rightarrow \infty} f_{n_k}(r_n) = \lim_{n \rightarrow \infty} \lim_{k \rightarrow \infty} f_{n_k}(r_n) = \lim_{k \rightarrow \infty} f_{n_k}(x)$$

$\uparrow$ conti $\uparrow$ prop. of $f(x)$ $\uparrow$ def. of $\hat{f}$ $\uparrow$ def. of $\hat{f}$ $\uparrow$ positive convergent double sequence. $\uparrow$ f_{n_k} monotone

And f has only countably many discontinuities, because it is monotone, by definition.

Now, we can construct a convergent subsequence of $\{f_{n_k}\}$ in such points, proved. $\square$
(specifically, redefine f using such subsequences.)

7.15.

무엇을 묻고 싶은 것이지?

7.16. Let $\{f_n\}$ be an equicontinuous sequence of functions on a compact set K , and $\{f_n\}$ converges pointwise. Then, $\{f_n\}$ converges uniformly.

Proof. Given $\epsilon > 0$, there exists $\delta > 0$ s.t.

$$d(x, y) < \delta, n \in \mathbb{N} \Rightarrow d(f_n(x), f_n(y)) \leq \epsilon. \quad (*)$$

Since K is compact, $\{B_{\frac{\delta}{2}}(x) \mid x \in K\}$ has a finite subcover, denote as $\{B_{\frac{\delta}{2}}(x_i) \mid i=1, \dots, n\}$

Since $f_n \rightarrow f$ pointwise, for each $i=1, \dots, n$, there is $N_i \in \mathbb{N}$ s.t.

$$n \geq N_i \Rightarrow d(f_n(x_i), f(x_i)) < \epsilon$$

Now, take $N = \max\{N_1, \dots, N_n\}$. Then, for all $i=1, \dots, n$,

$$n \geq N \Rightarrow d(f_n(x_i), f(x_i)) < \epsilon.$$

Let $x \in K$. Then, for some $i=1, \dots, n$, $x \in B_{\frac{\delta}{2}}(x_i)$. Then,

$$d(f_n(x), f(x)) \leq d(f_n(x), f_n(x_i)) + d(f_n(x_i), f(x_i)) + \underbrace{d(f(x_i), f(x))}_{\text{taking limit to } (*)} \leq 3\epsilon.$$

7.18. Let $\{f_n\}$ be a uniformly bounded sequence of functions, $f_n \in \mathcal{R}$ on $[a, b]$.

Define $F_n(x) = \int_a^x f_n(t) dt \quad (a \leq x \leq b)$

Then, $\{F_n\}$ has a uniformly convergent subsequence.
Proof.

17. 20. If f is continuous on $[0,1]$ and

$$\int_0^1 f(x) \cdot x^n dx = 0 \quad \text{for all } n=0,1,\dots$$

Then $f=0$.

sol). By the Weierstrass Thm, there exists a sequence of polynomials $\{p_n\}$ s.t. $p_n \rightarrow f$.

And, since f and p_n are continuous on the compact set $[0,1]$, they are bounded.

Therefore $f \cdot p_n \rightarrow f \cdot f = f^2$. Now, clearly $f \cdot p_n \in \mathcal{P}$, thus

$$\lim_{n \rightarrow \infty} \int_0^1 f(x) \cdot p_n(x) dx = \int_0^1 \lim_{n \rightarrow \infty} f(x) p_n(x) dx = \int_0^1 [f(x)]^2 dx$$

But by the assumption, for each $n \in \mathbb{N}$, $\int_0^1 f(x) p_n(x) dx = 0$.

Therefore, $\int_0^1 [f(x)]^2 dx = 0$, thus $f^2 = 0$, by $f^2 \geq 0$. Consequently $f=0$.